

# Zuhaib Akhtar

[u2024691@giki.edu.pk](mailto:u2024691@giki.edu.pk) | [linkedin.com/in/iZuhaibAkhtar](https://www.linkedin.com/in/iZuhaibAkhtar) | [github.com/ZuhaibAkhtarKhan](https://github.com/ZuhaibAkhtarKhan)

## EDUCATION

### Ghulam Ishaq Khan Institute of Technology (GIKI)

Topi, Pakistan

*Bachelor of Engineering in Software Engineering*

*Sep 2024 – Jul 2028*

- **RELEVANT COURSEWORK:** Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Database Management (SQL), Software Requirements Engineering, Computer Organization and Architecture, DevOps.

## EXPERIENCE

### Google Developers Group on Campus (GDGoC) GIKI

Dec 2024 – Present

*Core Team Member*

*Topi, Pakistan*

- Pioneered industry-academia integration initiatives for 100+ students, bridging technical knowledge gaps through official Google community support.
- Brokered 10+ industry-led speaker sessions featuring firms like Vyro.ai and Vector.ai, yielded internships for 10+ students via networking sessions.
- Executed 3 high-scale hackathons sponsored by GitHub, Base Technologies and Vyro's ImagineArt, managing 20+ teams and amassed USD 5000 in prize pools.

### Microsoft Learn Student Ambassadors Club GIKI

Nov 2024 – Present

*Core Team and Dev Team Member*

*Topi, Pakistan*

- Coordinated 5+ technical seminars and 2 hackathons with partners like CursorAI and Devsinc, securing PKR 200K+ in competitive prize pools.

## PROJECTS

### MiniJira (Project Management Software) | *PostgreSQL, Flask, React.js*

Apr 2026 – Present

- Architecting a full-stack Jira clone, specializing in **PostgreSQL** schema design to manage user authentication, ticket states, and role-based permissions.
- Optimizing data retrieval for a target base of 100+ entries, utilizing strategic indexing and SQL joins to ensure performant state management.
- Synchronizing in a 4-person team to mimic enterprise-level Scrum workflows and project tracking functionalities.

### KhataV2 (Voice-First Ledger) | *Gemini, MongoDB Atlas, Next.js 16, Twilio*

Apr 2026

- Engineered a voice-first financial ledger using Google Gemini LLMs to process Urdu/English natural language commands into structured JSON.
- Architected a high-performance system with Zustand for zero-latency, offline-ready tracking, benchmarked for 5,000+ potential daily users.
- Integrated Twilio WhatsApp API to automate receipts and summaries, mitigating literacy-based friction for small business owners.

### 3Bins (AI Smart Waste Separator) | *Python, YOLOv8, Arduino Mega, C++*

Feb 2026

- Developed an AI-powered smart waste separator utilizing a custom **YOLOv8** model, achieving a **0.60 mAP (60%)** for real-time classification.
- Implemented serial communication between model outputs and an Arduino Mega to automate physical sorting via a multi-servo mechanism.

### MedAi (Disease Diagnosis System) | *C++, DSA (BST, HashMap, Linked List)*

Oct 2025 – Nov 2025

- Engineered a diagnostic engine achieving  $O(\log N)$  search efficiency for  $N$  diseases by architecting a hybrid **Binary Search Tree** and **HashMap** structure.
- Optimized symptom-to-disease mapping to  $O(1)$  average-case complexity, significantly bypassing processing latency for real-time diagnostic queries.
- Utilized dynamic symptom storage through Linked Lists, ensuring memory-efficient data management for 100+ medical profiles.

## SKILLS

**Languages:** Python, C/C++, SQL (PostgreSQL), JavaScript, Tailwind CSS, HTML/CSS.

**Core Competencies:** Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Relational Databases, SDLC, DevOps.

**Developer Tools:** Git/GitHub, Docker, Linux, Jira, VS Code, CursorAI, Arduino IDE.

**Libraries/Frameworks:** LangChain, LangGraph, RAG Pipelines, Flask, Node.js, NumPy, Pandas.